`GeoModels` Tutorial: simulation, estimation and prediction of spatial data with heavy tails using t random fields

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Introduction

In this tutorial we show how to analyze spatial data with heavy tails using t random fields (Bevilacqua et al., 2021) with the `R` package `GeoModels` (Bevilacqua et al., 2026). The t distribution is a flexible parametric model, which is able to accommodate flexible tail behaviour and, in particular, heavier tails than the ones induced by Gaussian random fields.

We first load the `R` libraries needed in this tutorial and set the name of the model in the `GeoModels` package.

```
rm(list=ls())
require(GeoModels)
require(fields)
require(hypergeo)
model="StudentT" # model name in the GeoModels package
set.seed(199)
```

Simulation of t random fields

The definition of a t random field starts by considering a 'parent' Gaussian random field $G = \{G(\boldsymbol{s}), \boldsymbol{s} \in S\}$, where $\boldsymbol{s}$ represents a location in the domain S. In this tutorial we consider the spatial case i.e. $S \subseteq \mathbb{R}^2$. However, the package `GeoModels` allows to work also with spatio-temporal data or data defined on a sphere of arbitrary radius. The Gaussian field G is assumed weakly stationary with zero mean, unit variance and correlation function $\rho(\boldsymbol{h}) = \mathrm{cor}\{G(\boldsymbol{s}+\boldsymbol{h}), G(\boldsymbol{s})\}$.

Given $G_1, \ldots, G_\nu$ independent copies of G, also independent of G, where ν is a positive integer greater than two, let $Y_\nu^* = \{Y_\nu^*(\boldsymbol{s}), \boldsymbol{s} \in S\}$ be a random field defined through a scale mixture:

$$Y_\nu^*(\boldsymbol{s}) = \left(\sum_{i=1}^{\nu} G_i(\boldsymbol{s})^2 / \nu \right)^{-1/2} G(\boldsymbol{s}), \tag{1}$$

with marginal distribution t with associated density:

$$f_{Y_\nu^*(\boldsymbol{s})}(y) = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\,\Gamma(\nu/2)} \left(1 + \frac{y^2}{\nu}\right)^{-(\nu+1)/2}, \quad y \in \mathbb{R}. \tag{2}$$

Then $\mathrm{E}\{Y_\nu^*(\boldsymbol{s})\} = 0$, $\mathrm{var}\{Y_\nu^*(\boldsymbol{s})\} = \nu/(\nu-2)$, $\nu > 2$, and the correlation function is given

by Bevilacqua et al. (2021):

$$\rho_{Y_\nu^*}(\boldsymbol{h}) = \frac{(\nu-2)\Gamma^2\left(\frac{\nu-1}{2}\right)}{2\Gamma^2\left(\frac{\nu}{2}\right)} {}_2F_1\left(\frac{1}{2},\frac{1}{2};\frac{\nu}{2};\rho^2(\boldsymbol{h})\right)\rho(\boldsymbol{h}). \tag{3}$$

Here ${}_2F_1(a,b;c;x)$ is the Gaussian hypergeometric function (Abramowitz and Stegun, 1970). In the `GeoModels` package the ${}_2F_1$ function is computed using the function `hypergeo` of the `hypergeo` package (Hankin, 2016).

Then we define the location-scale transformation process $Y_\nu = \{Y_\nu(\boldsymbol{s}), \boldsymbol{s} \in S\}$ as:

$$Y_\nu(\boldsymbol{s}) := \mu(\boldsymbol{s}) + \sigma Y_\nu^*(\boldsymbol{s}), \tag{4}$$

with $\mathrm{E}\{Y_\nu(\boldsymbol{s})\} = \mu(\boldsymbol{s})$ and $\mathrm{var}\{Y_\nu(\boldsymbol{s})\} = \sigma^2\nu/(\nu-2)$, and a spatial regression model can be specified by assuming that $\mu(\boldsymbol{s}) = X(\boldsymbol{s})^T\boldsymbol{\beta}$ where $X(\boldsymbol{s})$ is a k-dimensional vector of covariates and $\boldsymbol{\beta} = (\beta_1,\ldots,\beta_k)^T$ is a k-dimensional vector of (unknown) parameters. In this tutorial we assume $k = 2$.

To obtain a simulation from Y_ν we need to specify regression mean, degrees of freedom and variance parameters i.e. $\beta_1, \beta_2, \nu, \sigma^2$ respectively. Additionally, we need to specify a parametric correlation $\rho(\boldsymbol{h})$ for the 'parent' Gaussian random field. We first set the spatial coordinates:

```
N=600
coords=cbind(runif(N),runif(N))
plot(coords,pch=20,xlab="",ylab="")
```

For the correlation function $\rho(\boldsymbol{h})$ of the 'parent' Gaussian random field G we assume an isotropic Matérn model (Matérn, 1986):

$$\rho_{\alpha,\gamma}(\boldsymbol{h}) = \frac{2^{1-\gamma}}{\Gamma(\gamma)}\left(\frac{\|\boldsymbol{h}\|}{\alpha}\right)^\gamma K_\gamma\left(\frac{\|\boldsymbol{h}\|}{\alpha}\right), \qquad \|\boldsymbol{h}\| \geq 0, \tag{5}$$

where K_γ is a modified Bessel function of the second kind of order γ, $\gamma > 0$ is the smoothness parameter and $\alpha > 0$ the spatial scale parameter. Then, we set the parameters associated with this correlation model:

```
corrmodel="Matern" ## correlation model
scale=0.2/3 ## scale parameter
smooth=0.5 ## smooth parameter
nugget=0 ## nugget parameter
```

and we set the degrees of freedom and scale parameter of the t random field:

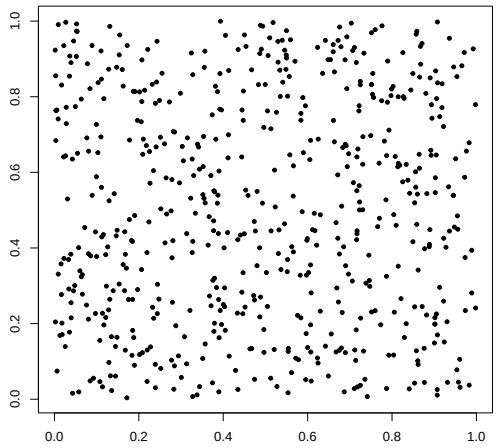


Figure 1: Spatial location sites used in the tutorial.

```
df=5; # degrees of freedom
sill=1; # sigma^2 (marginal variance = sill*df/(df-2))
```

Finally we set the mean regression parameters and the regression matrix:

```
mean=0.5; mean1=-1 # regression parameters
a0=rep(1,N); a1=runif(N,-1,1)
X=cbind(a0,a1) ## regression matrix
```

We are now ready to simulate a realization of the t random field Y_ν using the function `GeoSim`. Simulation is performed exploiting the stochastic representation (1), where the Gaussian fields involved are generated through Cholesky decomposition:

```
param=list(nugget=nugget,mean=mean,mean1=mean1,scale=scale,
          smooth=smooth,sill=sill,df=1/df)
data <- GeoSim(coordx=coords,corrmodel=corrmodel,
               param=param,model=model,X=X)$data
```

Note that the parametrization in the package `GeoModels` uses the inverse of the degrees of freedom, as suggested in Bevilacqua et al. (2021).

Estimation of t random fields

Estimation of regression, degrees of freedom and correlation parameters of the t random field Y_ν can be performed using pairwise likelihood estimation. Let $f_{Y^*_\nu;ij}(y_i, y_j)$ be the density of the bivariate random vector $(Y^*_\nu(\boldsymbol{s}_i), Y^*_\nu(\boldsymbol{s}_j))^T$ given by Bevilacqua et al. (2021):

$$f_{Y^*_\nu;ij}(y_i,y_j) = \frac{\nu^\nu l_{ij}^{-(\nu+1)/2}\Gamma^2((\nu+1)/2)}{\pi\Gamma^2(\nu/2)(1-\rho^2(\boldsymbol{h}))^{-(\nu+1)/2}} F_4\left(\frac{\nu+1}{2},\frac{\nu+1}{2};\frac{1}{2},\frac{\nu}{2};\frac{\nu\rho^2(\boldsymbol{h})y_i^2y_j^2}{l_{ij}},\frac{\nu^2\rho^2(\boldsymbol{h})}{l_{ij}}\right) + \frac{\rho(\boldsymbol{h})y_iy_j\nu^{\nu+2}l_{ij}^{-\nu/2-1}}{2\pi(1-\rho^2(\boldsymbol{h}))^{-(\nu+1)/2}} F_4\left(\frac{\nu}{2}+1,\frac{\nu}{2}+1;\frac{3}{2},\frac{\nu}{2};\frac{\nu\rho^2(\boldsymbol{h})y_i^2y_j^2}{l_{ij}},\frac{\nu^2\rho^2(\boldsymbol{h})}{l_{ij}}\right), \tag{6}$$

where $l_{ij} = [(y_i^2+\nu)(y_j^2+\nu)]$ and

$$F_4(a,b;c,c';w,z) = \sum_{k=0}^{\infty}\sum_{m=0}^{\infty}\frac{(a)_{k+m}(b)_{k+m}w^kz^m}{k!m!(c)_k(c')_m}, \qquad \sqrt{|w|}+\sqrt{|z|}<1,$$

is the Appell function of the fourth type (Gradshteyn and Ryzhik, 2007).

Given a partial realization $(y(\boldsymbol{s}_1),\ldots,y(\boldsymbol{s}_N))^T$ of the t random process Y_ν defined in equation (4), the density of the bivariate random vector $(Y_\nu(\boldsymbol{s}_i), Y_\nu(\boldsymbol{s}_j))^T$ can be obtained from (6) as:

$$f_{Y_\nu;ij}(y(\boldsymbol{s}_i),y(\boldsymbol{s}_j)) = \frac{1}{\sigma^2} f_{Y^*_\nu;ij}\left(\frac{y(\boldsymbol{s}_i)-\mu(\boldsymbol{s}_i)}{\sigma},\frac{y(\boldsymbol{s}_j)-\mu(\boldsymbol{s}_j)}{\sigma}\right). \tag{7}$$

Then, the pairwise likelihood function is defined as:

$$pl(\boldsymbol{\theta}) = \sum_{i=1}^{N}\sum_{j\neq i}^{N}\log\left(f_{Y_\nu;ij}(y(\boldsymbol{s}_i),y(\boldsymbol{s}_j))\right)w_{ij}, \tag{8}$$

where w_{ij} are non-negative weights, not depending on $\boldsymbol{\theta}$.

An efficient way to specify the (non symmetric) weights from the computational and efficient viewpoint is based on neighborhoods:

$$w_{ij}(k) = \begin{cases} 1, & \boldsymbol{s}_i \in N_k(\boldsymbol{s}_j), \\ 0, & \text{otherwise}, \end{cases} \tag{9}$$

where $N_k(\boldsymbol{s}_l)$ is the set of the neighbors of order $k = 1, 2, \ldots$ of the point $\boldsymbol{s}_l$. The pairwise likelihood estimator $\widehat{\boldsymbol{\theta}}_{pl}$ is obtained maximizing (8) with respect to $\boldsymbol{\theta} = (\beta_1, \beta_2, \nu, \sigma^2, \alpha, \gamma)^T$. In the `GeoModels` package we can choose the fixed parameters and the parameters that can be estimated.

As argued in Bevilacqua et al. (2021), the degrees of freedom must be fixed to a positive integer value greater than two for this stochastic construction (in some special cases $\nu > 2$ without any restriction on the degrees of freedom parameter). If we assume ν unknown, the degrees of freedom can be fixed through a two-step estimation. In the first step, we estimate the parameters, including ν without any restriction to integers. Under the reparametrization of the degrees of freedom the parametric space is $(0, 0.5)$. Pairwise likelihood estimation can be performed using the function `GeoFit`. In this example we perform optimization of (8) using the function `nlminb` that allows box-constrained optimization using PORT routines. However other types of optimization algorithms available in R can be used (BFGS or Nelder-Mead for instance).

```
optimizer="nlminb"
fixed1<-list(nugget=nugget,smooth=smooth)
start1<-list(mean=mean,mean1=mean1,scale=scale,sill=sill,df=1/df)
I=Inf
lower1<-list(mean=-I,mean1=-I,scale=1e-6,sill=1e-6,df=1e-4)
upper1<-list(mean=I,mean1=I,scale=I,sill=I,df=0.5-1e-4)
fit1 <- GeoFit(data=data,coordx=coords,corrmodel=corrmodel,
        optimizer=optimizer,lower=lower1,upper=upper1,
        neighb=3,X=X,start=start1,fixed=fixed1,model=model)
```

Note that the option `neighb=3` sets the number of neighborhood k in the weight function (9), i.e. $k = 3$. To respect the construction with integer degrees of freedom we round the estimation of ν obtained at the first step and ensure $\nu \geq 3$:

```
DF=max(3L,as.integer(round(1/fit1$param$df)))
print(DF)
# Original tutorial output: [1] 5
```

In the original execution the rounded estimated value of ν matches the true value $\nu = 5$. Then we perform the second step estimation keeping fixed the degrees of freedom:

```
start<-list(mean=mean,mean1=mean1,scale=scale,sill=sill)
fixed<-list(nugget=nugget,df=1/DF,smooth=smooth)
lower<-list(mean=-I,mean1=-I,scale=1e-6,sill=1e-6)
upper<-list(mean=I,mean1=I,scale=I,sill=I)
fit2 <- GeoFit(data=data,coordx=coords,corrmodel=corrmodel,
        optimizer=optimizer,lower=lower,upper=upper,
```

```
        neighb=3,X=X,start=start,fixed=fixed,model=model)
```

The object `fit2` includes information about the pairwise likelihood estimation:

```
fit2
# Original tutorial (2024) output:
# Maximum Composite-Likelihood Fitting of StudentT Random Fields
# Setting: Marginal Composite-Likelihood
# Model: StudentT; Covariance model: Matern
# Type of convergence: Successful
# Maximum log-Composite-Likelihood value: -5308.37
# Estimated parameters:
# mean mean1 scale sill
# 0.28956 -1.01657 0.05267 0.91928
```

An alternative, less efficient and computationally easier estimator can be obtained by assuming a misspecified Gaussian model in the pairwise likelihood estimation method. Specifically, if in the estimation step we assume a Gaussian random field with the same mean, variance and correlation function of the t random field (see Bevilacqua et al. (2021)), then a weighted misspecified Gaussian pairwise likelihood estimation can be performed changing the name of the model in the function `GeoFit`:

```
fit3 <- GeoFit(data=data,coordx=coords,corrmodel=corrmodel,
        optimizer=optimizer,lower=lower,upper=upper,
        neighb=3,X=X,start=start,fixed=fixed,
        model="Gaussian_misp_StudentT")
```

The two estimates are quite similar in this case but in general the misspecified Gaussian assumption leads to a loss of efficiency that increases when decreasing the degrees of freedom.

```
unlist(fit2$param); unlist(fit3$param)
# Original tutorial (2024) outputs:
# mean mean1 scale sill
# 0.28956411 -1.01657485 0.05267308 0.91928083
# 0.3223977 -1.1058598 0.0537655 0.9345408
```

Checking model assumptions

Given the estimation of the mean regression and sill parameters, the estimated residuals

$$\widehat{Y_\nu^*(\boldsymbol{s}_i)} = \frac{y(\boldsymbol{s}_i) - X(\boldsymbol{s}_i)^T\widehat{\boldsymbol{\beta}}}{(\widehat{\sigma^2})^{1/2}}, \qquad i = 1, \ldots, N,$$

can be viewed as a realization of the process Y_ν^*. The residuals can be computed using the `GeoResiduals` function:

```
res=GeoResiduals(fit2) # computing residuals
```

The marginal distribution assumption on the residuals can be graphically checked through a QQ-plot using the `GeoQQ` function (see Figure 2, left part):

```
GeoQQ(res)
```

The covariance model assumption can be checked comparing the empirical and the estimated semi-variogram using the `GeoVariogram` and `GeoCovariogram` functions (see Figure 2, right part):

```
vario <- GeoVariogram(data=res$data,
                  coordx=coords,maxdist=0.4)
GeoCovariogram(res,show.vario=TRUE,
                  vario=vario,pch=20,ylim=c(0,2))
```

The semi-variogram is computed using the correlation function (3).

Prediction of t random fields

For a given spatial location $\boldsymbol{s}_0$ with associated covariates $X(\boldsymbol{s}_0)$, the optimal linear prediction of a t random field is given by:

$$\widehat{Y_\nu}(\boldsymbol{s}_0) = X(\boldsymbol{s}_0)^T\widehat{\boldsymbol{\beta}} + \sum_{i=1}^{N} \lambda_i \big[y(\boldsymbol{s}_i) - X(\boldsymbol{s}_i)^T\widehat{\boldsymbol{\beta}}\big], \tag{10}$$

where the vector of weights $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_N)^T$ is given by $\boldsymbol{\lambda} = R_\nu^{-1}\boldsymbol{c}_\nu$ and

- $\boldsymbol{c}_\nu = \big(\text{cor}(Y_\nu(\boldsymbol{s}_0), Y_\nu(\boldsymbol{s}_1)), \ldots, \text{cor}(Y_\nu(\boldsymbol{s}_0), Y_\nu(\boldsymbol{s}_N))\big)^T$.
- $R_\nu = [\text{cor}(Y_\nu(\boldsymbol{s}_i), Y_\nu(\boldsymbol{s}_j))]_{i,j=1}^N$ is the correlation matrix.

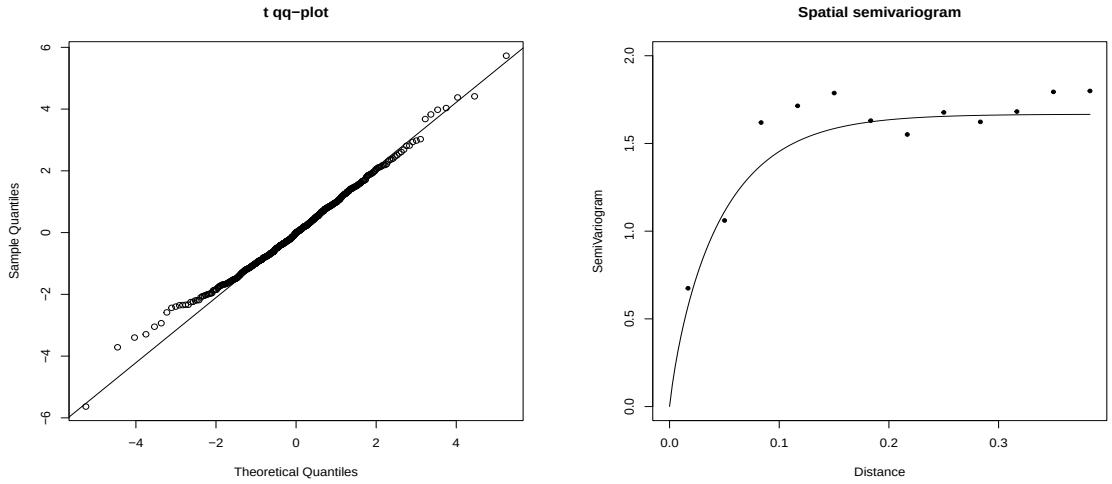


Figure 2: From left to right: QQ-plot of the residuals using the t_5 distribution and empirical vs estimated semi-variogram for the residuals (original tutorial).

Moreover the associated mean square error (MSE) is given by:

$$MSE(\widehat{Y_\nu}(\boldsymbol{s}_0)) = \sigma^2 \frac{\nu}{\nu - 2} \left(1 - \boldsymbol{c}_\nu^T R_\nu^{-1} \boldsymbol{c}_\nu\right). \tag{11}$$

The predictor can be viewed as an optimal Gaussian predictor assuming (3) as correlation function. If the parameters are unknown, both (10) and (11) can be computed replacing the parameters with the pairwise likelihood estimates. In particular, R_ν and $\boldsymbol{c}_\nu$ can be computed using (3) coupled with the estimates of the Matérn correlation function and of the degrees of freedom.

Kriging and associated MSE can be obtained using the `GeoKrig` function. We first need to specify the spatial locations to predict and, in this example, we consider a spatial regular grid:

```
xx=seq(0,1,0.012)
loc_to_pred=as.matrix(expand.grid(xx,xx))
Nloc=nrow(loc_to_pred)
Xloc=cbind(rep(1,Nloc),runif(Nloc))
```

Then the optimal linear prediction (10), using the final fitted parameters, can be performed using the `GeoKrig` function (computation can be time consuming):

```
param_est=as.list(c(fit2$param,fixed))
pr=GeoKrig(data=data,coordx=coords,loc=loc_to_pred,
```

```
        X=X,Xloc=Xloc,corrmodel=corrmodel,model=model,
        mse=TRUE,param=param_est)
```

Finally a kriging map with associated mean square error (Figure 3) can be obtained:

```
par(mfrow=c(1,3))
quilt.plot(coords[,1],coords[,2],data,main="Data")
map=matrix(pr$pred,ncol=length(xx))
image.plot(xx,xx,map,xlab="",ylab="",main="SimpleKriging")
map_mse=matrix(pr$mse,ncol=length(xx))
image.plot(xx,xx,map_mse,xlab="",ylab="",main="MSE")
```

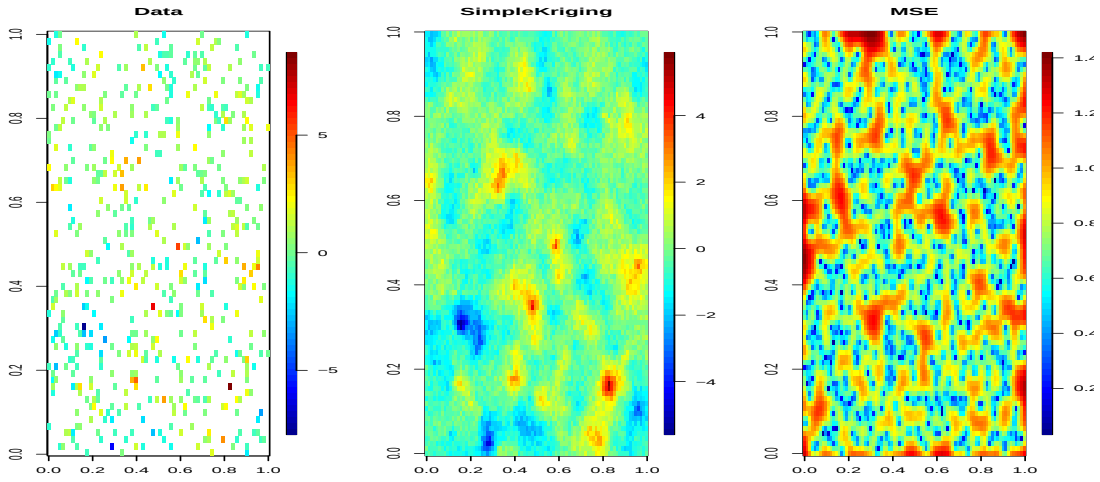


Figure 3: From left to right: observed spatial data, associated kriging map and mean square error map (original tutorial, before correcting the prediction parameter list).

References

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